ERASURE ENTROPIES OF SPARSE LINEAR DNFS

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Abstract. For a Boolean function f and a coordinate i, the erasure entropy $E_i(f)$ is the conditional entropy of the event $i \in U$ given $U \setminus \{i\}$, where U is drawn from the spectral distribution $\hat{f}^2$. The Fourier entropy-influence conjecture is equivalent to every non-constant f having a coordinate with $E_i(f) = O(\mathrm{Inf}_i(f))$. We consider monotone DNFs whose terms all have width w, any two of which share at most one variable, and in which every variable lies in at most D terms. We show that $E_i(f)/\mathrm{Inf}_i(f) = 2 + o(1)$ at every relevant coordinate, uniformly in the number of variables and terms, as $w \to \infty$ with $D \le 2^{(1-\eta)(w-1)}$ for a fixed $\eta > 0$, and we give an explicit error bound for each w, D and degree of i. The proof rests on a separation inequality valid for every monotone function: if the derivative $D_i f$ puts little spectral weight outside a family Ω of sets and the restriction $f|_{x_i=0}$ puts little weight on Ω, then E_i/Inf_i is close to 2. For a linear DNF, Ω is the family of sets that contain many of the other variables of some term through i.

1. Introduction

For $f\colon \{0,1\}^n \to \{-1,1\}$ write $f = \sum_S \hat{f}(S)\chi_S$ with $\chi_S(x) = (-1)^{\sum_{j\in S} x_j}$, and let U be a random subset with $\Pr[U = S] = \hat{f}(S)^2$. The Fourier entropy is $H(f) = H(U)$ and the total influence is $I(f) = \mathbb{E}|U| = \sum_i \mathrm{Inf}_i(f)$, where $\mathrm{Inf}_i(f) = \Pr[i \in U]$. The Fourier entropy-influence (FEI) conjecture of Friedgut and Kalai [3] asks for an absolute C with $H(f) \le C\, I(f)$ for every Boolean f. It is known for symmetric functions [9], for read-once formulas over gates of bounded arity [8] and over AND and OR gates [2], and for read-k decision trees with k fixed [13]; its min-entropy version is known for read-k DNFs with k fixed [10, 1], and Han [5] bounds $H(f)$ by a multiple of $I(f)$ plus $\sum_i \mathrm{Inf}_i(f)\log_2(1/\mathrm{Inf}_i(f))$. Kelman, Kindler, Lifshitz, Minzer and Safra [7] prove a weaker form of the conjecture: the spectrum of a Boolean function of constant variance is concentrated on $2^{O(I(f)\log I(f))}$ characters.

The erasure entropy (in the sense of Verdú and Weissman [12]) of f at i is

$$E_i(f) = H\big(\mathbf{1}[i \in U] \,\big|\, U \setminus \{i\}\big).$$

It is the part of the entropy drop from f to its two restrictions at x_i that is not controlled by a constant times Inf_i [5, Remarks 3.1–3.2], [4, (32)], and the FEI conjecture holds if and only if there is an absolute c such that every non-constant Boolean function has a coordinate with $E_i(f) \le c\,\mathrm{Inf}_i(f)$ [11]. The trivial bound $E_i \le h_2(\mathrm{Inf}_i)$ loses a factor $\log(1/\mathrm{Inf}_i)$; the question is when that factor can be removed, and at which coordinates.

This paper computes the ratio E_i/Inf_i asymptotically for a class of DNFs in which the coordinates are linked by many terms. Let F be a monotone DNF on $\{0,1\}^n$. We call F *linear* if any two of its terms share at most one variable, *w-uniform* if all its terms have w variables, and of *degree at most D* if every variable lies in at most D terms. Put $f = 1 - 2F$. Read-once DNFs are the linear DNFs of degree one; a linear DNF of degree D is read-D, but its terms may overlap in a pattern with many cycles, for instance the m rows and m columns of an $m \times m$ array.

Theorem 1. *Fix $\eta \in (0,1)$. Over all n, all w-uniform linear monotone DNFs F on $\{0,1\}^n$ of degree at most $D \le 2^{(1-\eta)(w-1)}$, and all coordinates i with $\mathrm{Inf}_i(f) > 0$,*

$$\sup\Big|\frac{E_i(f)}{\mathrm{Inf}_i(f)} - 2\Big| \longrightarrow 0 \qquad (w \to \infty).$$

In particular $\sum_i E_i(f) = (2+o(1))\, I(f)$ on this class.

Theorem 1 follows from an explicit inequality, Theorem 6, valid for each width, degree and coordinate. The value 2 is the limit of E_i/Inf_i for a coordinate inside one long AND that nothing else touches: for an AND of $k+1$ variables a direct computation gives $E_i/\mathrm{Inf}_i = 2(1-2^{-k}) + 2^k \Phi\big((1-2^{-k})^2, 4^{-k}\big)$ at each of them (with Φ as in Section 2), which differs from 2 by at most $2^{-k}(2k+2)$. Theorem 1 says that in a sparse linear DNF every coordinate behaves like that, however many terms pass through it and however the terms around it overlap.

The proof separates two spectra. Write g and h for the even and odd parts of f in x_i. For a monotone F, h is the indicator that i is pivotal and $g = 2N - 1 - h$, where N is the indicator that F vanishes at $x_i = 0$. If the spectrum of h lives on a family Ω of sets on which N has almost no weight, then on Ω the two coefficients $\hat{g}(T)$ and $\hat{h}(T)$ have nearly the same square, and each such T contributes nearly $2\hat{h}(T)^2$ to E_i. Proposition 2 makes this quantitative for every monotone function. For a linear DNF the family is Ω_k, the sets that contain at least k of the other variables of some term through i: the pivotal indicator spreads its spectral weight evenly over the subsets of the other variables of each such term (Lemma 4), while the background event N needs k further terms to reach k of these variables, which costs $((D-1)2^{-(w-1)})^k$ (Lemma 5).

Theorem 1 is a statement about the ratio at every coordinate and does not prove the FEI conjecture for the class. The equivalence of [11] needs a good coordinate in every restriction of f, and restrictions of a uniform linear DNF have terms of different widths.

2. A separation inequality

Fix i and write $f^{(b)}$ for the restriction $x_i = b$, a function of the other coordinates. Put

$$g = \tfrac{1}{2}\big(f^{(0)} + f^{(1)}\big), \qquad h = \tfrac{1}{2}\big(f^{(0)} - f^{(1)}\big).$$

Then $\hat{g}(T) = \hat{f}(T)$ and $\hat{h}(T) = \hat{f}(T \cup \{i\})$ for $T \subseteq [n] \setminus \{i\}$, and $\mathbb{E}h^2 = \mathrm{Inf}_i(f) =: \delta$. Put $p(T) = \hat{g}(T)^2$ and $q(T) = \hat{h}(T)^2$, and

$$\Phi(a,b) = \eta(a) + \eta(b) - \eta(a+b), \qquad \eta(x) = -x\log_2 x.$$

Since $(U \setminus \{i\}, \mathbf{1}[i \in U])$ takes the value $(T,0)$ with probability $p(T)$ and $(T,1)$ with probability $q(T)$, we have $E_i(f) = \sum_T \Phi(p(T), q(T))$. The function Φ is concave, homogeneous of degree one, increasing in each argument, and $\Phi(b,b) = 2b$. Writing $\Phi(a,b) = a\log_2(1+b/a) + b\log_2(1+a/b)$ and using $\ln(1+t) \le t$ gives

$$\Phi(a,b) \le b\log_2\Big(e\,\frac{a+b}{b}\Big). \tag{1}$$

Now let F be monotone increasing and $f = 1-2F$. Put $N = (1+f^{(0)})/2$ and $M = (1+f^{(1)})/2$, the indicators that F vanishes at $x_i = 0$ and at $x_i = 1$. Then $M \le N$, $h = N - M$ is the indicator that i is pivotal, $\hat{h}(\emptyset) = \mathbb{E}h = \delta$, and

$$g = N + M - 1 = 2N - 1 - h, \qquad \hat{g}(T) = 2\hat{N}(T) - \hat{h}(T) \quad (T \neq \emptyset).$$

Proposition 2. *Let F be monotone, $f = 1-2F$, $\delta = \mathrm{Inf}_i(f) > 0$, and let N be as above. Let Ω be a family of non-empty subsets of $[n] \setminus \{i\}$ and let $\varepsilon \in (0,1]$ and $\kappa \ge 0$ satisfy*

$$\sum_{T \notin \Omega} \hat{h}(T)^2 \le \varepsilon\delta, \qquad \sum_{T \in \Omega} \hat{N}(T)^2 \le \kappa\delta.$$

Then

$$2-2\varepsilon-\frac{8}{\ln 2}\sqrt{\kappa}\ \le\ \frac{E_i(f)}{\delta}\ \le\ 2+\frac{8}{\ln 2}\sqrt{\kappa}+\varepsilon\log_2\Big(e\Big[3+\frac{8\mathrm{Var}N}{\varepsilon\delta}\Big]\Big)+\delta\log_2\big(e[1+\delta^{-2}]\big).$$

Proof. Split the sum $E_i=\sum_T\Phi(p(T),q(T))$ into $T\in\Omega$, $T\notin\Omega$ with $T\neq\emptyset$, and $T=\emptyset$.

On Ω. Fix $b\ge 0$. The function $u\mapsto\Phi(u^2,b^2)$ is even, equals $2b^2$ at $u=\pm b$, and for $u\neq 0$ has derivative $2u\log_2(1+b^2/u^2)$, whose absolute value is at most $(4/\ln 2)b$ because $\ln(1+t^2)\le 2t$ for $t\ge 0$. For $T\in\Omega$ take $b=|\hat h(T)|$ and move u from $-\hat h(T)$ to $\hat g(T)=2\hat N(T)-\hat h(T)$, a distance $2|\hat N(T)|$:

$$\big|\Phi(p(T),q(T))-2q(T)\big|\le\frac{8}{\ln 2}|\hat h(T)||\hat N(T)|.$$

By Cauchy–Schwarz, $\sum_\Omega|\hat h||\hat N|\le\sqrt{\delta}\sqrt{\kappa\delta}$, and $(1-\varepsilon)\delta\le\sum_\Omega q\le\delta$. Hence

$$2(1-\varepsilon)\delta-\frac{8}{\ln 2}\delta\sqrt{\kappa}\le\sum_{T\in\Omega}\Phi(p(T),q(T))\le 2\delta+\frac{8}{\ln 2}\delta\sqrt{\kappa}.$$

Since every term of E_i is non-negative, this gives the lower bound.

Off Ω. Let P and Q be the sums of p and q over the non-empty $T\notin\Omega$; if $Q=0$ these terms vanish, so let $Q>0$. Then $Q\le\varepsilon\delta$, and from $p=(2\hat N-\hat h)^2\le 8\hat N^2+2\hat h^2$ we get $P\le 8\mathrm{Var}N+2\varepsilon\delta$. By (1) and Jensen's inequality for the concave function $t\mapsto\log_2 t$,

$$\sum\Phi(p,q)\le\sum q\log_2\Big(e\,\frac{p+q}{q}\Big)\le Q\log_2\Big(e\,\frac{P+Q}{Q}\Big)\le Q\log_2\Big(e\,\frac{c}{Q}\Big),\qquad c=8\mathrm{Var}N+3\varepsilon\delta.$$

The last expression increases in Q on $(0,c]$, so it is at most $\varepsilon\delta\log_2(e[3+8\mathrm{Var}N/(\varepsilon\delta)])$.

At $T=\emptyset$, $q=\delta^2$ and $p\le 1$, so (1) gives $\Phi\le\delta^2\log_2(e[1+\delta^{-2}])$. Adding the three parts gives the upper bound. □

3. Linear DNFs

From now on F is a w-uniform linear monotone DNF of degree at most D, $f=1-2F$, $L=w-1\ge 1$ and $a=2^{-L}$. Fix a coordinate i that lies in $d\ge 1$ terms $t_1,\dots,t_d$. The *mates* of t_s are its L variables other than i, and A_s is the indicator that all of them equal 1. By linearity the mate sets are pairwise disjoint. A *background term* is a term that does not contain i; N is the indicator that every background term fails, and

$$M=N\prod_{s=1}^{d}(1-A_s),\qquad h=N-M=N\cdot\mathbf{1}\big[\textstyle\bigvee_s A_s\big],\qquad\delta=\mathbb{E}h.$$

Put $A=\mathbb{E}N$.

Two facts about linear DNFs are used repeatedly. A background term meets the mate set of t_s in at most one variable, since it shares at most one variable with t_s and does not contain i. And a term through a mate of t_s, other than t_s itself, is a background term, since otherwise it would share two variables with t_s. We also use Harris's inequality [6]: under a product measure, an increasing and a decreasing event are negatively correlated, and two increasing events are positively correlated.

Lemma 3. *With $\theta=1-L(D-1)a-(d-1)a/2$,*

$$da\,\theta\,A\le\delta\le da\,A.$$

Proof. The events A_s are increasing and N is decreasing, so $\Pr[N,A_s]\le Aa$, and the union bound gives $\delta\le daA$. For the lower bound, let R_s be the indicator that every background term disjoint from the mate set of t_s fails, so that $N\le R_s$. The remaining background terms meet that mate set in exactly one variable; each mate lies in at most $D-1$ of them, so there are at

most $L(D-1)$, and given $A_s = 1$ each of them fires exactly when its other L variables, which are not mates of t_s, all equal 1. Let V_s be the event that one of these residual conjunctions equals 1; it is increasing and $\Pr[V_s] \le L(D-1)a$. Since R_s and V_s do not depend on the mates of t_s,

$$\Pr[N, A_s] = a \Pr[R_s, \overline{V_s}] \ge a\big(\Pr[R_s] - \Pr[R_s]\Pr[V_s]\big) \ge aA\big(1 - L(D-1)a\big),$$

using Harris for R_s and V_s and, in the last step, $\Pr[R_s] \ge A$; if $1 - L(D-1)a < 0$ the bound holds trivially. For $s \ne u$ the event $A_s A_u$ is increasing with probability a^2, so $\Pr[N, A_s, A_u] \le Aa^2$. Bonferroni's inequality gives $\delta \ge daA(1 - L(D-1)a) - \binom{d}{2}a^2 A = da\theta A$. □

For $1 \le k \le L$ let Ω_k be the family of sets $T \subseteq [n] \setminus \{i\}$ that contain at least k mates of some t_s, and put

$$\gamma_k = a \sum_{j<k} \binom{L}{j}.$$

Lemma 4. $\sum_{T \notin \Omega_k} \hat{h}(T)^2 \le \min(1, d\gamma_k)\, \delta$.

Proof. Put $h_s = NA_s \prod_{u<s}(1 - A_u)$. These functions have disjoint supports and sum to h, so $\sum_s \|h_s\|_2^2 = \delta$. Since h_s vanishes unless $A_s = 1$, $h_s = A_s \Psi_s$, where Ψ_s is $N \prod_{u<s}(1 - A_u)$ with the mates of t_s set to 1, a function of the variables outside that mate set. As $A_s = 2^{-L} \sum_{S \subseteq \text{mates}(t_s)} (-1)^{|S|} \chi_S$, every Fourier coefficient of h_s has the form $\hat{h}_s(S \cup V) = \pm 2^{-L} \hat{\Psi}_s(V)$ with S a set of mates and V disjoint from them. The weight of $\hat{h}_s^2$ on sets with exactly j mates of t_s is therefore $\binom{L}{j} 4^{-L} \|\Psi_s\|_2^2 = \binom{L}{j} a \|h_s\|_2^2$, using $\|h_s\|_2^2 = a\|\Psi_s\|_2^2$. A set outside Ω_k has fewer than k mates of t_s, so $\sum_{T \notin \Omega_k} \hat{h}_s(T)^2 \le \gamma_k \|h_s\|_2^2$. By Cauchy–Schwarz,

$$\sum_{T \notin \Omega_k} \hat{h}(T)^2 \le d \sum_s \sum_{T \notin \Omega_k} \hat{h}_s(T)^2 \le d\gamma_k \delta.$$

The bound by δ is trivial. □

Lemma 5. *If $k(D-1)a < 2$, then*

$$\sum_{T \in \Omega_k} \hat{N}(T)^2 \le d\, \alpha_k A, \qquad \alpha_k = \binom{L}{k} \frac{2^{\binom{k}{2} - 2k} \big((D-1)a\big)^k}{1 - k(D-1)a/2}.$$

Proof. Every $T \in \Omega_k$ contains a set J of k mates of some t_s, and there are $d\binom{L}{k}$ such sets J. For each of them, $\sum_{T \supseteq J} \hat{N}(T)^2 = \|D_J N\|_2^2$, where D_J is the composition of the derivatives $D_j N = \frac{1}{2}(N|_{x_j=0} - N|_{x_j=1})$ over $j \in J$. So it suffices to show $\|D_J N\|_2^2 \le \alpha_k \binom{L}{k}^{-1} A$.

Fix J. A background term contains at most one element of J. Let R' be the indicator that every background term avoiding J fails, and for $j \in J$ let Q_j be the OR of the residual conjunctions, of width L, of the at most $D-1$ background terms through j. Neither R' nor any Q_j depends on the variables in J, and

$$N = R' \prod_{j \in J} (1 - x_j Q_j), \qquad D_J N = \pm R' \prod_{j \in J} \tfrac{1}{2} Q_j, \qquad \|D_J N\|_2^2 = 4^{-k} \Pr[R', \textstyle\bigwedge_j Q_j].$$

The event $\bigwedge_j Q_j$ is increasing and R' decreasing, so $\Pr[R', \bigwedge_j Q_j] \le \Pr[R'] \Pr[\bigwedge_j Q_j]$. For $\bigwedge_j Q_j$ one residual conjunction must equal 1 for each j; there are at most $(D-1)^k$ choices, the chosen terms are distinct because each contains exactly one element of J, and two residual

conjunctions share at most one variable, so their union has at least $kL - \binom{k}{2}$ variables. Hence $\Pr[\bigwedge_j Q_j] \le (D-1)^k 2^{\binom{k}{2}} a^k$. Finally, x_j is independent of R' and Q_j, so by Harris

$$A \ge \Pr[R'] - \sum_{j\in J} \Pr[R', x_j Q_j] \ge \Pr[R']\Big(1 - \sum_j \tfrac{1}{2}\Pr[Q_j]\Big) \ge \Pr[R']\big(1 - \tfrac{1}{2}k(D-1)a\big).$$

Combining the three displays gives $\|D_J N\|_2^2 \le 4^{-k}(D-1)^k 2^{\binom{k}{2}} a^k A/(1-k(D-1)a/2)$. □

Theorem 6. *Let F be a w-uniform linear monotone DNF of degree at most D, $L = w-1$, $a = 2^{-L}$, and let i lie in $d \ge 1$ terms. Let $1 \le k \le L$, and put*

$$\theta = 1 - L(D-1)a - \frac{(d-1)a}{2}, \qquad \varepsilon = \min(1, d\gamma_k), \qquad \kappa = \frac{\alpha_k}{a\theta},$$

with γ_k and α_k as above. If $\theta > 0$, $k(D-1)a < 2$ and $da \le 1/4$, then

$$\Big|\frac{E_i(f)}{\mathrm{Inf}_i(f)} - 2\Big| \le 2\varepsilon + \frac{8}{\ln 2}\sqrt{\kappa} + \varepsilon \log_2\Big(e\Big[3 + \frac{8}{\varepsilon\, da\, \theta}\Big]\Big) + da \log_2\big(e[1 + (da)^{-2}]\big).$$

Proof. Apply Proposition 2 with $\Omega = \Omega_k$, which consists of non-empty sets. Lemma 4 gives the first hypothesis with ε (note $\varepsilon \ge da > 0$). Lemmas 5 and 3 give $\sum_{\Omega_k} \hat{N}^2 \le d\alpha_k A \le \alpha_k \delta/(a\theta) = \kappa\delta$. In the upper bound, $\mathrm{Var} N \le \mathbb{E}N = A \le \delta/(da\theta)$, and $\delta \le daA \le da \le 1/4$; the function $x \mapsto x\log_2(e[1+x^{-2}])$ is increasing on $(0, 1/4]$, since its derivative is $\log_2(1+x^{-2}) + \log_2 e\,(1 - 2/(1+x^2)) \ge \log_2 17 - \log_2 e > 0$ there. □

Proof of Theorem 1. Let $k = \lfloor 1/\eta \rfloor + 1$, so $k\eta > 1$, and let $L \to \infty$ with $D \le 2^{(1-\eta)L}$. Then $Da \le 2^{-\eta L}$, so $L(D-1)a \to 0$, $\theta \to 1$, $k(D-1)a < 2$ and $da \le Da \le 1/4$ for L large, uniformly in $d \le D$. Next $d\gamma_k \le Da\, kL^k \le kL^k 2^{-\eta L} \to 0$, and

$$\kappa \le \frac{2L^k 2^{\binom{k}{2}} (Da)^k}{a\theta} \le \frac{2L^k 2^{\binom{k}{2}}}{\theta}\, 2^{(1-k\eta)L} \to 0.$$

In the third term, $x \mapsto x\log_2(e[3 + c/x])$ is increasing, $\varepsilon \le kL^k 2^{-\eta L}$, and $c = 8/(da\theta) \le 16 \cdot 2^L$ for L large, so the term is $O(kL^{k+1}2^{-\eta L})$. The last term is at most $2^{-\eta L}\log_2(e[1+4^{\eta L}]) = O(L 2^{-\eta L})$. None of these bounds depends on n, on the number of terms or on i. □

Remark 7. The two error sources pull k in opposite directions. Since $\gamma_k \le kL^{k-1}a$, the weight of h off Ω_k is small once $DkL^{k-1} \ll 2^L$, which favours small k; the weight of N on Ω_k is small once $L^k D^k \ll 2^{(k-1)L}$ up to factors depending on k, which favours large k. With $D = 2^{\beta L}$, $\beta < 1$, both hold for $k > 1/(1-\beta)$, and this is where the hypothesis $D \le 2^{(1-\eta)L}$ comes from.

4. Numerical checks

All values below were computed with exact Walsh–Hadamard transforms. The script `check_lin.py` in the code repository evaluates the four inequalities of Lemmas 3–5 and Theorem 6 at every coordinate and every admissible k for 17 linear DNFs: stars of D terms through one variable, the rows and columns of a 4×4 array, and twelve random linear DNFs with $w \in \{4, 5\}$, $D \in \{2, 3, 4\}$ and 16 to 18 variables. All 784 (coordinate, k) pairs satisfy all four inequalities, and the left side of Theorem 6 is never more than 0.14 of the right side; at these widths the bound is far from tight.

Table 1 shows the ratio approaching 2 as the width grows, for the star of D terms of width w through a common variable, at the centre ($d = D$) and at another variable ($d = 1$). For the rows and columns of an $m \times m$ array the ratio is 2.629 at $m = 3$ and 3.245 at $m = 4$.

D	w	n	Inf centre	E/Inf centre	Inf other	E/Inf other
2	4	7	0.2344	2.685	0.1094	2.909
2	6	11	0.0615	2.450	0.0303	2.351
2	8	15	0.0156	2.178	0.0078	2.118
2	10	19	0.0039	2.060	0.0019	2.037
3	5	13	0.1760	2.692	0.0549	2.666
3	7	19	0.0461	2.378	0.0151	2.225
4	5	17	0.2275	2.690	0.0515	2.728

TABLE 1. E_i/Inf_i for the star of D terms of width w on $n = 1 + D(w-1)$ variables.

Acknowledgements and funding

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [14] and working together inside the Brain framework [16], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [15], the Math Brain repository [14] and the Brain framework repository [16].

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [16] and Math Brain [14]. In November 2026 a Claude Fable agent (Anthropic) and Codex agents running OpenAI's GPT-6.1, connected to the Brain and working on the same problem, studied erasure entropies of DNFs; a Codex agent found the theorem for sparse linear DNFs and its proof. Claude Opus 5.5 (Anthropic) rederived each proof, wrote the numerical checks, prepared the manuscript and audited it. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Code and outputs: `https://github.com/yspennstate/fei-sparse-linear-dnf`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [14].

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